

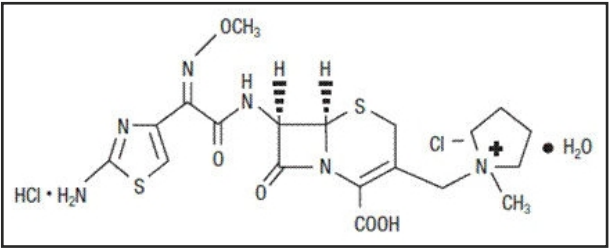


ANNEXURE - I  
ART WORK APPROVAL FORM

(Ref. SOP. No.: DFD080)

For the use of a Registered Medical Practitioner only  
**CEFEPIME FOR INJECTION USP**  
**CEFICAD (For IM/IV use only)**

**Each vial contains:**  
Cefepime Hydrochloride USP  
Equivalent to Cefepime 1000mg  
Sterile mixture of Cefepime Hydrochloride and L-Arginine



**DESCRIPTION**

**Ceficad Injection** (cefepime hydrochloride) is a semi-synthetic, broad spectrum, cephalosporin antibiotic for parenteral administration. The chemical name is 1-[[[6R, 7R)-7-[2-[2-amino-4-thiazolyl)-glyoxylamino]-2-carboxy-8-oxo-5-thia-1-azabicyclo[4.2.0]oct-2-en-3-yl]methyl]-1-methylpyrrolidiniumchloride, 7Z-(Z)-(O-methylloxime), monohydrochloride.

**Cefepime Hydrochloride**  
Cefepime hydrochloride contains the equivalent of not less than 825 µg and not more than 911 µg of cefepime (C<sub>16</sub>H<sub>18</sub>N<sub>4</sub>O<sub>5</sub>S<sub>2</sub>) per mg, calculated on an anhydrous basis. It is highly soluble in water.

**Ceficad 1000 Inj:** White to pale yellow powder contained in 20 ml flint type III glass vials sealed with 20 mm grey butyl rubber stopper and 20mm pink coloured flip off aluminium seal.

**Description of after reconstitution:**  
Fresh constituted solutions of Ceficad (Cefepime Hydrochloride) will range in color from colorless to amber.

**PHARMACOLOGICAL PROPERTIES**

**Pharmacokinetic Properties**  
Adults: Average plasma concentrations of cefepime observed in normal adult males at various times following single 30-minute infusions or intramuscular injections of 500mg, 1g and 2g are summarized.

Following intramuscular administration, cefepime is completely absorbed.

Cefepime is metabolised to N-methylpyrrolidine, which is rapidly converted to the N-oxide. Urinary recovery of unchanged cefepime accounts for approximately 85% of the administered dose; high concentrations of unchanged cefepime are found in the urine. Less than 1% of the administered dose is recovered from urine as N-methylpyrrolidine, 6.8% as the N-oxide, and 2.5% as an epimer of cefepime. Serum protein binding of cefepime averages 16.4% and is independent of concentration in the serum.

The average elimination half-life of cefepime is approximately 2 hours, and does not vary with respect to dose over the range of 250mg to 2g. There was no accumulation in healthy subjects receiving doses up to 2g intravenously every 8 hours for a period of 9 days. Total body clearance averages 120mL/min. The average renal clearance of cefepime is 110mL/min, suggesting that cefepime is eliminated almost exclusively by renal mechanisms, primarily glomerular filtration.

**Pharmacodynamic Properties**  
Cefepime hydrochloride is a semi-synthetic broad-spectrum cephalosporin antibiotic for parenteral administration. Cefepime is a bactericidal agent that acts by inhibition of bacterial wall synthesis. It has a broad spectrum of activity against a wide range of gram-positive and gram-negative bacteria including most strains resistant to aminoglycosides or third-generation cephalosporins such as ceftazidime. Cefepime is highly resistant to hydrolysis by most beta-lactamases, has a low affinity for chromosomally-encoded beta-lactamases, and exhibits rapid penetration into gram-negative bacterial cells.

In studies using *Escherichia coli* and *Enterobacter cloacae*, cefepime bound with highest affinity to penicillin binding protein (PBP) 3 followed by PBP 2, then PBPs 1a and 1b. Binding to PBP 2 occurs with significantly higher affinity than that of other parenteral cephalosporins. This may enhance its antibacterial activity. The moderate affinity of cefepime for PBPs 1a and 1b probably also contributes to its overall bactericidal activity. Cefepime has been shown to be bactericidal by time-kill analysis (killing-curves) and by determination of minimum bactericidal concentrations (MBC) for a wide variety of bacteria. The cefepime MBC/MIC ratio was 2 for more than 80% of isolates of all gram-positive and gram-negative species tested. Synergy with aminoglycosides has been demonstrated *in vitro*, primary with *Pseudomonas aeruginosa* isolates. Cefepime has been shown to be active against most strains of the following organisms:

**Antibacterial Spectrum**

**Gram-positive Aerobes:** *Staphylococcus aureus* (including beta-lactamase-producing strains) *Staphylococcus epidermidis* (including beta-lactamase-producing strains) Other staphylococci including *S. hominis*, *S. saprophyticus*, *Streptococcus pyogenes* (Group A streptococci), *Streptococcus agalactiae* (Group B streptococci), *Streptococcus pneumoniae* (including intermediate penicillin resistant strains with penicillin MIC of 0.1 to 1 mcg/ml) Other - *Haemolytic streptococci* (Group C, G, F), *S. bovis* (Group D), *Viridans streptococci*.

**Note:** Most strains of enterococci, e.g: *Enterococcus faecalis*, and methicillin-resistant staphylococci are resistant to most cephalosporins including cefepime.

**Gram-negative Aerobes:**  
*Pseudomonas* sp. including *P. aeruginosa*, *P. putida*, *P. stutzeri*, *Escherichia coli*, *Klebsiella* sp. including *K. pneumoniae*, *K. oxytoca*, *K. ozaenae*, *Enterobacter* sp. including *E. cloacae*, *E. aerogenes*, *E. agglomerans*, *E. sakazakii*, *Proteus* sp. including *P. mirabilis*, *P. vulgaris*, *Acinetobacter calcoaceticus* (subsp. anitratus, lwofii), *Aeromonas hydrophila*, *Capnocytophaga* sp., *Citrobacter* sp. including *C. diversus*, *C. freundii*, *Campylobacter jejuni*, *Gardnerella vaginalis*, *Haemophilus ducreyi*, *Haemophilus influenzae* (including beta-lactamase-producing strains), *Haemophilus parainfluenzae*, *Hafnia alvei*, *Legionella* sp., *Morganella morganii*, *Moraxella catarrhalis* (Branhamella catarrhalis) (including beta-lactamase-producing strains) *Neisseria gonorrhoeae* (including beta-lactamase-producing strains), *Neisseria meningitidis* *Providencia* sp. including *P. rettgeri*, *P. stuartii*, *Salmonella* sp., *Serratia* including *S. marcescens*, *S. liquefaciens*, *Shigella* sp., *Yersinia enterocolitica*

**Note:** Cefepime is inactive against many strains of *Stenotrophomonas maltophilia* (formerly known as *Xanthomonas maltophilia*) and *Pseudomonas maltophilia*.

**Anaerobes:**  
*Bacteroides* sp., *Clostridium perfringens*, *Fusobacterium* sp., *Mobiluncus* sp., *Peptostreptococcus* sp., *Prevotella melaninogenica* (formerly known as *Bacteroides melaninogenicus*), *Veillonella* sp.

**Note:** Cefepime is inactive against *Bacteroides fragilis* and *Clostridium difficile*.

**Indication**  
Ceficad Injection (Cefepime Hydrochloride) is indicated in the treatment of the following infections caused by susceptible strains of the designated microorganisms: Pneumonia (moderate to severe) caused by *Streptococcus pneumoniae*, including cases associated with concurrent bacteremia, *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, or *Enterobacter species*.

**Empiric Therapy for Febrile Neutropenic Patients**  
Cefepime as monotherapy is indicated for empiric treatment of febrile neutropenic patients. In patients at high risk for severe infection (including

patients with a history of recent bone marrow transplantation, with hypotension at presentation, with an underlying hematologic malignancy, or with severe or prolonged neutropenia), antimicrobial monotherapy may not be appropriate. Insufficient data exist to support the efficacy of cefepime monotherapy in such patients.

**Urinary Tract Infections**  
Uncomplicated and Complicated Urinary Tract Infections (including pyelonephritis) caused by *Escherichia coli* or *Klebsiella pneumoniae*, when the infection is severe, or caused by *Escherichia coli*, *Klebsiella pneumoniae*, or *Proteus mirabilis*, when the infection is mild to moderate, including cases associated with concurrent bacteremia with these microorganisms.

**Skin and Skin Structure Infections**  
Uncomplicated Skin and Skin Structure Infections caused by *Staphylococcus aureus* (methicillin susceptible strains only) or *Streptococcus pyogenes*.

**Intra-abdominal Infections**  
Complicated Intra-abdominal Infections caused by *Escherichia coli*, *Pseudomonas aeruginosa*, or *Bacteroides fragilis*.

To reduce the development of drug-resistant bacteria and maintain the effectiveness of cefepime hydrochloride and other antibacterial drugs, cefepime hydrochloride should be used only to treat or prevent infections that are proven or strongly suspected to be caused by susceptible bacteria. When culture and susceptibility information are available, they should be considered in selecting or modifying antibacterial therapy. In the absence of such data, local epidemiology and susceptibility patterns may contribute to the empiric selection of therapy.

**DOSAGE AND METHOD OF ADMINISTRATION**  
The recommended adult and pediatric dosages and routes of administration via intravenous or intramuscular are outlined in the following **table 1**. Ceficad should be administered intravenously over approximately 30 minutes.

**Table 1: Recommended Dosage Schedule for Ceficad in Patients with CrCL>60mL/min**

| Site and Type of Infection                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Dose            | Frequency | Duration (days) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|-----------------|
| <b>Adults</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |           |                 |
| <b>Moderate to Severe</b> Pneumonia due to <i>S.pneumoniae</i> *, <i>P. aeruginosa</i> , <i>K. pneumoniae</i> , or <i>Enterobacter</i> species                                                                                                                                                                                                                                                                                                                                                        | 1-2 g IV        | q12h      | 10              |
| Empiric therapy for febrile neutropenic patients                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2 g IV          | q8h       | 7**             |
| <b>Mild to Moderate</b> Uncomplicated or Complicated Urinary Tract Infections, including pyelonephritis, due to <i>E. coli</i> , <i>K. pneumoniae</i> or <i>P.mirabilis</i> *                                                                                                                                                                                                                                                                                                                         | 0.5-1g IV/IM*** | q12h      | 7-10            |
| <b>Severe</b> Uncomplicated or Complicated Urinary Tract Infections, including pyelonephritis, due to <i>E. coli</i> or <i>K. pneumonia</i> *                                                                                                                                                                                                                                                                                                                                                         | 2 g IV          | q12h      | 10              |
| <b>Mild to Moderate</b> Uncomplicated Skin and Skin Structure Infections                                                                                                                                                                                                                                                                                                                                                                                                                              | 1 g IV          | q12h      | 10              |
| <b>Complicated Intra-abdominal Infections</b> caused by <i>E. coli</i> , <i>P. aeruginosa</i> or <i>B. fragilis</i> .                                                                                                                                                                                                                                                                                                                                                                                 | 2 g IV          | q12h      | 7-10            |
| <b>Pediatric patients (2 months up to 12 years)</b><br>The maximum dose for pediatric patients should not exceed the recommended adult dose. The usual recommended dosage in pediatric patients up to 40 kg in weight for uncomplicated and complicated urinary tract infections (including pyelonephritis), uncomplicated skin and skin structure infections, and pneumonia is 50 mg/kg/dose, administered q12h (50 mg/kg/dose, q8h for febrile neutropenic patients), for durations as given above. |                 |           |                 |

\* Including cases associated with concurrent bacteremia.  
\*\* or until resolution of neutropenia. In patients whose fever resolves but who remain neutropenic for more than 7 days, the need for continued antimicrobial therapy should be re-evaluated frequently.  
\*\*\* IM route of administration is indicated only for mild to moderate, uncomplicated or complicated UTIs due to *E. coli* when the IM route is considered to be a more appropriate route of drug administration.  
**Elderly** – Dose adjustment is not required, unless there is concurrent renal impairment.

**Impaired Hepatic Function** – No adjustment is necessary for patients with impaired hepatic function.

**Impaired Renal Function** – In patients with impaired renal function (creatinine clearance ≤ 60 mL/min), the dose of Ceficad Injection (Cefepime Hydrochloride) should be adjusted to compensate for the slower rate of renal elimination. The recommended initial dose of Ceficad Injection (Cefepime Hydrochloride) should be the same as in patients with normal renal function except in patients undergoing hemodialysis. The recommended maintenance doses of Ceficad Injection (Cefepime Hydrochloride) in patients with renal insufficiency are presented in Table 2.

**Table 2 Maintenance Dosing Schedule for Ceficad in Adult Patients (Normal Renal Function, Renal Insufficiency, and Hemodialysis)**

| Creatinine Clearance (mL/min) | Recommended Maintenance Dosage      |                                                     |                       |                       |
|-------------------------------|-------------------------------------|-----------------------------------------------------|-----------------------|-----------------------|
| >60                           | Usual dose, no adjustment necessary |                                                     |                       |                       |
|                               | 2 g every 8 hours                   | 2 g every 12 hours                                  | 1 g every 12 hours    | 500 mg every hours    |
| 30-60                         | 2 g every 12 hours                  | 2 g every 24 hours                                  | 1 g every 24 hours    | 500 mg 24 hours       |
| 11-29                         | 2 g every 24 hours                  | 1 g every 24 hours                                  | 500 mg every 24 hours | 500 mg every 24 hours |
| <11                           | 1 g every 24 hours                  | 500 mg every 24 hours                               | 250 mg every 24 hours | 250 mg every 24 hours |
| CAPD                          | 2 g every 48 hours                  | 2 g every 48 hours                                  | 1 g every 48 hours    | 500 mg every 48 hours |
| Hemodialysis*                 | 1 g every 24 hours                  | 1 g on day 1, then 500 mg every 24 hours thereafter |                       |                       |

\*On hemodialysis days, cefepime should be administered following hemodialysis. Whenever possible, cefepime should be administered at the same time each day.

When only serum creatinine is available, the following formula (Cockcroft and Gault equation) may be used to estimate creatinine clearance. The serum creatinine should represent a steady state of renal function:

**Males:** Creatinine Clearance (mL/min) =  $\frac{\text{Weight (kg)} \times (140 - \text{age})}{72 \times \text{serum creatinine (mg/dL)}}$   
**Females:** 0.85 × above value

**Intravenous infusion:** Constitute the 1 or 2 g piggy back (100 ml) bottle with 50 or 100 ml of a compatible IV fluid. Intermittent IV infusion with a Y type administration set can be accomplished with compatible solutions. However, during infusion of a solution containing cefepime, it is desirable to discontinue the other solution.

CADILA PHARMACEUTICALS LTD.

Product Ceficad 1000

Export (Malaysia)

Dimension 180x424 mm

Item code Oxxxx

Mfg. Location Dholka

Component Leaflet-Front

Change part Layout: NA

Substrate: 40 gsm bible paper printed pack insert (Folding size: 90 x 53 mm)

Superseded item code No. : OEI0029

Color : Pantone P. Black C

No. of. Colors : 01

Pharma Code : 00

Version : 02

Reason for issue : Text matter changed

Approved by

|               |            |       |            |      |
|---------------|------------|-------|------------|------|
| Sign & Date : |            |       |            |      |
| Name :        |            |       |            |      |
| Designation : |            |       |            |      |
| Department :  | Pack F & D | F & D | Regulatory | Q.C. |
|               |            |       | Production | Q.A. |

Size: 180 x 424 mm-Front Side

P P Black C

HV-08/10/16-V02

HV-16/09/16-V01





ANNEXURE - I  
ART WORK APPROVAL FORM

(Ref. SOP. No.: DFD080)

**Intramuscular Administration:** For IM administration, CEFICAD (Cefepime HCl) should be constituted with one of the following diluents: sterile water for injection, 0.9% sodium chloride, 5% dextrose injection, 0.5% or 1.0% lidocaine hydrochloride, or sterile bacteriostatic water for injection with parabens or benzyl alcohol.

**Dialysis Patients**  
In patients undergoing continuous ambulatory peritoneal dialysis, Cefepime Hydrochloride may be administered at normally recommended doses at a dosage interval of every 48 hours.  
In patients undergoing hemodialysis, approximately 68% of the total amount of cefepime present in the body at the start of dialysis will be removed during a 3-hour dialysis period. The dosage of Cefepime Hydrochloride for hemodialysis patients is 1 g on Day 1 followed by 500 mg every 24 hours for the treatment of all infections except febrile neutropenia, which is 1 g every 24 hours.  
Cefepime Hydrochloride should be administered at the same time each day following the completion of hemodialysis on hemodialysis days.

**Children with Impaired Renal Function**  
Data in pediatric patients with impaired renal function are not available; however, since cefepime pharmacokinetics are similar in adults and pediatric patients, changes in the dosing regimen proportional to those in adults (**see Tables 1 and 2**) are recommended for pediatric patients.

**Preparation of solution and Administration**  
**CEFICAD** (Cefepime HCl) is to be constituted using the volumes of diluent shown in Table 3; the diluents to be used are identified following this table.

Table 3 Preparations of solution of Ceficad

|                          | Amount of diluent to be added (mL) | Approximate available volume (mL) | Approximate cefepime concentration (mg/mL) |
|--------------------------|------------------------------------|-----------------------------------|--------------------------------------------|
| Intravenous<br>1g vial   | 10                                 | 11.4                              | 90                                         |
| Intramuscular<br>1g vial | 3.0                                | 4.4                               | 230                                        |

**COMPATIBILITY AND STABILITY**  
**Intravenous:**  
**CEFICAD** (Cefepime HCl) is compatible at concentrations between 1 mg/ml and 40 mg/ml with the following IV infusion fluids: 0.9% sodium chloride, 5% and 10% dextrose injection, M/6 sodium lactate injection, 5% dextrose and 0.9% sodium chloride injection, lactated ringers and 5% dextrose injection. These solutions may be stored up to 24 hours at controlled room temperature 20-25°C (68-77°F) or 7 days in a refrigerator 2-8°C (36-46°F). Solutions of cefepime HCl, like those of most beta-lactam antibiotics, should not be added to solution of ampicillin at a concentration greater than 40mg/ml, and should not added to metronidazole, vancomycin, gentamicin, tobramycin sulfate or netilmicin sulfate because of potential interaction. However, if concurrent therapy with **CEFICAD** (Cefepime HCl) is indicated, each of these antibiotics can be administered separately.

**Intramuscular:**  
**CEFICAD** (Cefepime HCl) constituted as directed is stable for 24 hours at controlled room temperature 20-25°C (68-77°F) or for 7 days in a refrigerator 2-8°C (36-46°F) with the following diluents: Sterile Water for Injection, 0.9% Sodium Chloride Injection, 5% Dextrose Injection, Sterile Bacteriostatic Water for Injection with Parabens or Benzyl Alcohol or 0.5% or 1% Lidocaine Hydrochloride.  
**NOTE: PARENTERAL DRUGS SHOULD BE INSPECTED VISUALLY FOR PARTICULATE MATTER BEFORE ADMINISTRATION AND NOT USED IF PARTICULATE MATTER IS PRESENT.**

As with other cephalosporins, the color of Ceficad powder, as well as its solutions, tend to darken depending on storage conditions; however, the product potency is not adversely affected.

**CONTRAINDICATIONS**  
CEFEPIME HYDROCHLORIDE is contraindicated in patients who have had previous hypersensitivity reactions to any component of the formulation, the cephalosporin class of antibiotics, penicillins or other beta-lactam antibiotics.

**WARNINGS**  
BEFORE THERAPY WITH CEFEPIME HYDROCHLORIDE FOR INJECTION IS INSTITUTED, CAREFUL INQUIRY SHOULD BE MADE TO DETERMINE WHETHER THE PATIENT HAS HAD PREVIOUS IMMEDIATE HYPERSENSITIVITY REACTIONS TO CEFEPIME, CEPHALOSPORINS, PENICILLINS, OR OTHER DRUGS. IF THIS PRODUCT IS TO BE GIVEN TO PENICILLIN-SENSITIVE PATIENTS, CAUTION SHOULD BE EXERCISED BECAUSE CROSS-HYPERSENSITIVITY AMONG BETA-LACTAM ANTIBIOTICS HAS BEEN CLEARLY DOCUMENTED AND MAY OCCUR IN UP TO 10% OF PATIENTS WITH A HISTORY OF PENICILLIN ALLERGY. IF AN ALLERGIC REACTION TO CEFEPIME HYDROCHLORIDE OCCURS, DISCONTINUE THE DRUG. SERIOUS ACUTE HYPERSENSITIVITY REACTIONS MAY REQUIRE TREATMENT WITH EPINEPHRINE AND OTHER EMERGENCY MEASURES INCLUDING OXYGEN, CORTICOSTEROIDS, INTRAVENOUS FLUIDS, INTRAVENOUS ANTIHISTAMINES, PRESSOR AMINES, AND AIRWAY MANAGEMENT, AS CLINICALLY INDICATED.

In patients with impaired renal function (creatinine clearance ≤ 60 mL/min), the dose of cefepime hydrochloride should be adjusted to compensate for the slower rate of renal elimination. Because high and prolonged serum antibiotic concentrations can occur from usual dosages in patients with renal insufficiency or other conditions that may compromise renal function, the maintenance dosage should be reduced when cefepime is administered to such patients. Degree of renal impairment, severity of infection, and susceptibility of the causative organisms should determine continued dosage. During postmarketing surveillance, serious adverse events have been reported including life-threatening or fatal occurrences of the following : encephalopathy (disturbance of consciousness including confusions, hallucinations, stupor, and coma), myoclonus, and seizures. Most cases occurred in patients with renal impairment who received doses of cefepime that exceeded the recommended dosage schedules. However, some cases of encephalopathy occurred in patients receiving a dosage adjustment for their renal function. In the majority of cases, symptoms of neurotoxicity were reversible and resolved after discontinuation of cefepime and/or after hemodialysis. Pseudomembranous colitis has been reported with nearly all-antibacterial agents, including cefepime hydrochloride, and may range in severity from mild to life threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhea subsequent to the administration of antibacterial agents. Treatment with antibacterial agents alters the normal flora of the colon and may permit overgrowth of clostridia. Studies indicate that a toxin produced by *Clostridium difficile* is a primary cause of "antibiotic-associated colitis".

After the diagnosis of pseudomembranous colitis has been established, therapeutic measures should be initiated. Mild cases of pseudomembranous colitis usually respond to drug discontinuation alone. In moderate-to-severe cases, consideration should be given to management with fluids and electrolytes, proteins supplementation, and treatment with an antibacterial drug clinically effective against *Clostridium difficile* colitis.

PRECAUTIONS

**General**  
Prescribing cefepime hydrochloride in the absence of proven or strongly suspected bacterial infection or a prophylactic indication is unlikely to provide benefit to the patients and increases the risk of the development of drug-resistant bacteria. As with other antimicrobials, prolonged use of cefepime hydrochloride may result in overgrowth of nonsusceptible microorganisms. Repeated evaluation of the patient's condition is essential. Should superinfection occur during therapy, appropriate measures should be taken.

Many cephalosporins, including cefepime, have been associated with a fall in prothrombin activity. Those at risk include patients with renal or hepatic impairment, or poor nutritional state, as well as patients receiving a protracted course of antimicrobial therapy. Prothrombin time should be monitored in patients with renal or hepatic impairment, or poor nutritional state, as well as patients receiving a protracted course of antimicrobial therapy. Prothrombin time should be monitored in patients at risk, and exogenous vitamin K administered as indicated. Positive direct Coombs' tests have been reported during treatment with cefepime hydrochloride. In hematologic studies or in transfusion cross-matching procedures when antiglobulin tests are performed on the minor side or in Coombs' testing of newborns whose mothers have received cephalosporin antibiotics before parturition, it should be recognized that a positive Coombs' test might be due to the drug. Cefepime hydrochloride should be prescribed with caution in individuals with a history of gastrointestinal disease, particularly colitis. Arginine has been shown to alter glucose metabolism and elevate serum potassium transiently when administered at 33 times the amount provided by the maximum recommended human dose of cefepime hydrochloride. The effect of lower doses is not presently known.

**Geriatric Use**  
When geriatric patients received the usual recommended adult dose, clinical efficacy and safety were comparable to clinical efficacy and safety in nongeriatric adult patients. Serious adverse events have occurred in geriatric patients with renal insufficiency given unadjusted doses of cefepime, including life-threatening or fatal occurrences of the following: encephalopathy, myoclonus, and seizures. This drug is known to be substantially excreted by the kidney, and the risk of toxic reactions to this drug may be greater in patients with impaired renal function.

Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and renal function should be monitored.

**Pregnancy**  
**Teratogenic effects-Pregnancy Category B**  
Cefepime was not teratogenic or embryocidal when administered during the period of organogenesis to rats at doses up to 1000 mg/kg/day (4 times the recommended maximum human dose calculated on a mg/m<sup>2</sup> basis) or to mice at doses up to 1200 mg/kg (2 times the recommended maximum human dose calculated on a mg/m<sup>2</sup> basis) or to rabbits at a dose level of 100 mg/kg (approximately equal to the recommended maximum human dose calculated on a mg/m<sup>2</sup> basis). There are, however, no adequate and well-controlled studies of cefepime use in pregnant women. Because animal reproduction studies are not always predictive of human response, this drug should be used during pregnancy only if clearly needed.

**Labor and Delivery**  
Cefepime has not been studied for use during labor and delivery. Treatment should only be given if clearly indicated.

**Lactation**  
Cefepime is excreted in human breast milk in very low concentrations (0.5 µg/mL). Caution should be exercised when cefepime is administered to a nursing woman.

Pediatric Use

The safety and effectiveness of cefepime in the treatment of uncomplicated and complicated urinary tract infections (including pyelonephritis), uncomplicated skin and skin structure infections, pneumonia, and as empiric therapy for febrile neutropenic patients have been established in the age groups 2 months up to 16 years. Use of cefepime hydrochloride in these age groups is supported by evidence from adequate and well controlled studies of cefepime in adults with additional pharmacokinetic and safety data from pediatric trials. Safety and effectiveness in pediatric patients below the age of 2 months have not been established.

There are insufficient clinical data to support the use of cefepime hydrochloride in pediatric patients under 2 months of age or for the treatment of serious infections in the pediatric population where the suspected or proven pathogen is *Haemophilus influenzae* type b.

**IN THOSE PATIENTS IN WHOM MENINGEAL SEEDING FROM A DISTANT INFECTION SITE OR IN WHOM MENINGITIS IS SUSPECTED OR DOCUMENTED, AN ALTERNATE AGENT WITH DEMONSTRATED CLINICAL EFFICACY IN THIS SETTING SHOULD BE USED.**

**ADVERSE REACTIONS**  
Cefepime Hydrochloride is generally well tolerated. Adverse events considered to be of definite, probable, or possible relationship to Cefepime Hydrochloride are listed below.

Hypersensitivity – rash, pruritus, urticaria.  
Gastrointestinal – nausea, vomiting, oral moniliasis, diarrhea, colitis  
Central nervous system – headache, vasodilation, dyspnea, dizziness, paraesthesia.

Local reactions at the site of IV infusion occurred in patients; these included phlebitis and inflammation.

**INTERACTION WITH OTHER MEDICATIONS AND OTHER FORMS OF INTERACTIONS:**

Renal function should be monitored carefully if high doses of aminoglycosides are to be administered with cefepime hydrochloride because of the increased potential of nephrotoxicity and ototoxicity of aminoglycoside antibiotics. Nephrotoxicity has been reported following concomitant administration of other cephalosporins with potent diuretics such as furosemide.

**INCOMPATIBILITY:**  
Cefepime should not be added to Ampicillin solution, metronidazole, vancomycin, gentamicin, tobramycin sulfate or netilmicin sulfate.

**OVERDOSAGE**  
Patients who received an overdose should be carefully observed and given supportive treatment. In the presence of renal insufficiency, hemodialysis, not peritoneal dialysis, is recommended to aid in the removal of cefepime from the body. Accidental overdosing has occurred when large doses were given to patients with impaired renal function. Symptoms of overdose include encephalopathy (disturbance of consciousness including confusion, hallucinations, stupor, and coma), myoclonus, seizures, and neuromuscular excitability.

**HOW SUPPLIED**  
Ceficad (Cefepime Hydrochloride) for injection is supplied for intermuscular or intravenous administration in strengths equivalent to 1g of Cefepime. Ceficad (Cefepime Hydrochloride) is a sterile, dry mixture of Cefepime Hydrochloride and L-arginine. The L-arginine, at an approximate concentration of 725 mg/g of cefepime, is added to control the pH of the constituted solution at 4.0-6.0. Freshly constituted solutions of Ceficad (Cefepime Hydrochloride) will range in color from colorless to amber.

**STORAGE:**  
Store below 30°C, protected from light and moisture. Keep all medicine out of reach of children. The reconstitution solution for IV and IM can be stored up to 24 hours at controlled room temperature 20-25°C or 7 days in a refrigerator 2-8°C.

**PRESENTATION:**  
**Ceficad 1000 Inj:** Unit Pack of 20 ml

Revision Date: February 2012

Manufactured by:  
  
1389, Dholka - 382225, INDIA.

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CADILA PHARMACEUTICALS LTD.

Product Ceficad 1000

Market Export (Malaysia)

Dimension 180x424 mm

Item code Oxxxx

Mfg. Location Dholka

Component Leaflet-Back

Change part Layout: NA

Substrate: 40 gsm bible paper printed pack insert (Folding size: 90 x 53 mm)

Superseded item code No. : OEI0029

Color : Pantone P. Black C

No. of. Colors : 01

Pharma Code : 00

Version : 02

Reason for issue : Text matter changed

Approved by

|               |            |       |            |      |            |
|---------------|------------|-------|------------|------|------------|
| Sign & Date : |            |       |            |      |            |
| Name :        |            |       |            |      |            |
| Designation : |            |       |            |      |            |
| Department :  | Pack F & D | F & D | Regulatory | Q.C. | Production |
|               |            |       |            |      | Q.A.       |

Size: 180 x 424 mm-Back Side

P P Black C

HV-08/10/16-V02  
HV-16/09/16-V01